

SIMATS ENGINEERING

ASSESSMENT TOOL 1: Class Test

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA1522

COURSE OUTCOME COVERED

CO4: *Analyze big data characteristics and challenges across domains including security and application aspects.*
(BL4)

Dave's Taxonomy: Articulation (Level 4)
Question Count: 10

Total Marks: 30

Code of Conduct

I **Atmakuri Sai Mrudula (192372087)** certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: **A. Sai Mrudula**

Assessment Tasks

Class Test

1. Analyze a multinational retail organization struggling with fragmented customer data across online and offline channels, and propose a unified big data architecture that enables real-time customer analytics and personalized recommendations.
2. Evaluate the effectiveness of distributed computing models in processing large-scale scientific research data, and justify how parallel processing improves computational efficiency and scalability compared to centralized systems.
3. Design a fraud detection framework for a digital payment platform using big data analytics, and justify how streaming data processing and predictive models improve fraud prevention accuracy.
4. Assess the challenges of storing and processing multimedia data such as video, audio, and images in a big data environment, and recommend suitable technologies and storage mechanisms for efficient management.
5. Construct a disaster recovery and business continuity strategy for a cloud-based big data platform, and justify how redundancy, replication, and automated failover mechanisms ensure operational resilience.
6. Analyze the impact of data quality issues such as inconsistency, duplication, and missing values in a large-scale analytics platform, and propose data governance strategies to improve analytical reliability.
7. Design a scalable recommendation system for an online streaming service using big data technologies, and justify how user behavior analytics and distributed processing enhance recommendation accuracy and user engagement.
8. Evaluate the role of cloud computing in modern big data ecosystems, and critically examine its advantages and limitations in terms of scalability, cost management, flexibility, and security.
9. Develop a real-time analytics framework for transportation and logistics management, and justify how big data technologies improve route optimization, fuel efficiency, and operational decision-making.

10. Critically analyze ethical and legal challenges associated with big data collection and analytics, and recommend organizational policies that ensure responsible data usage, regulatory compliance, and user trust.

Class Test Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Concept Accuracy	30%	Accurate and well-expressed technical answers	Mostly accurate answers	Partial accuracy	Inaccurate answers
Coverage	20%	Covers all required points	Covers most points	Limited coverage	Very poor coverage
Use of Examples	15%	Uses relevant and meaningful examples	Some relevant examples	Weak examples	No useful examples
Analytical Awareness	20%	Shows strong awareness of issues and implications	Reasonable awareness	Basic awareness	Minimal awareness
Clarity	15%	Clear and concise writing	Generally clear	Some unclear expressions	Poor clarity

1. Analyze a multinational retail organization struggling with fragmented customer data across online and offline channels, and propose a unified big data architecture that enables real-time customer analytics and personalized recommendations.

Ans:

A multinational retail company collects customer data from many sources such as online shopping websites, mobile apps, POS systems, loyalty cards and social media. Since the data is stored in different systems, it becomes difficult to understand the complete behaviour of a customer.

A unified big data architecture can combine all these sources. Data is first collected using data ingestion tools. Real-time data can be handled using technologies such as Apache Kafka. The collected data can be stored in a distributed data lake.

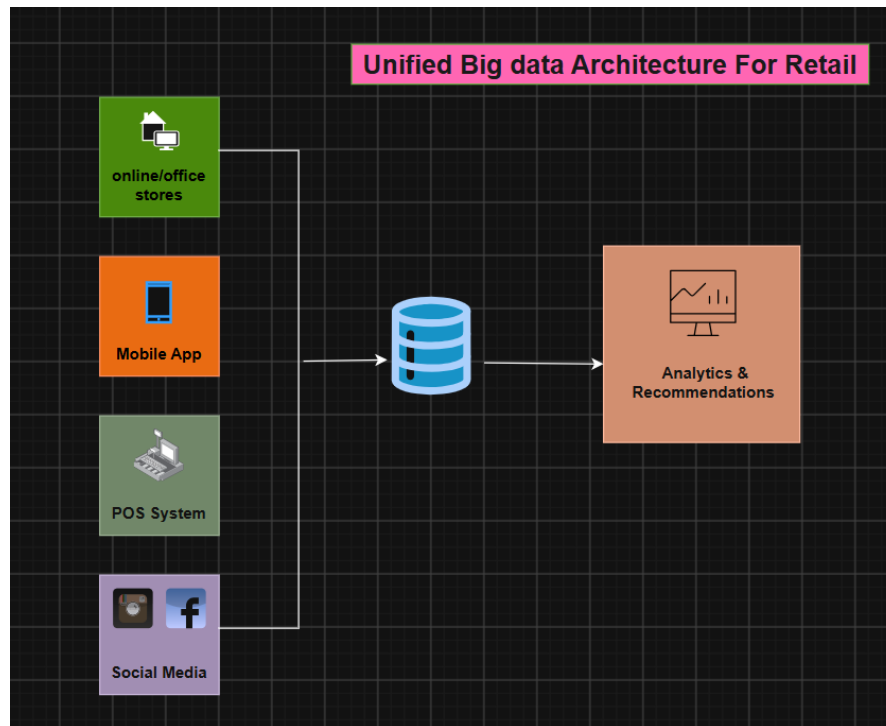
Tools such as Apache Spark can clean and process the data. Duplicate customer records can be removed and information from different sources can be combined. After processing, analytics can be used to understand customer behaviour.

Machine learning can then provide personalized recommendations. For example, if a customer regularly purchases sports shoes, the system can recommend sports clothes or new shoes.

The basic flow is:

Online + Offline Data → Data Ingestion → Data Lake → Spark Processing → Customer Analytics → Recommendations

This system provides a complete view of customers, improves personalized marketing and increases customer satisfaction and sales.



2. Evaluate the effectiveness of distributed computing models in processing large-scale scientific research data, and justify how parallel processing improves computational efficiency and scalability compared to centralized systems.

Ans:

Scientific research produces very large datasets from satellites, sensors, laboratories and simulations. Processing all this data using one computer can take a long time. Distributed computing solves this problem by using multiple computers together.

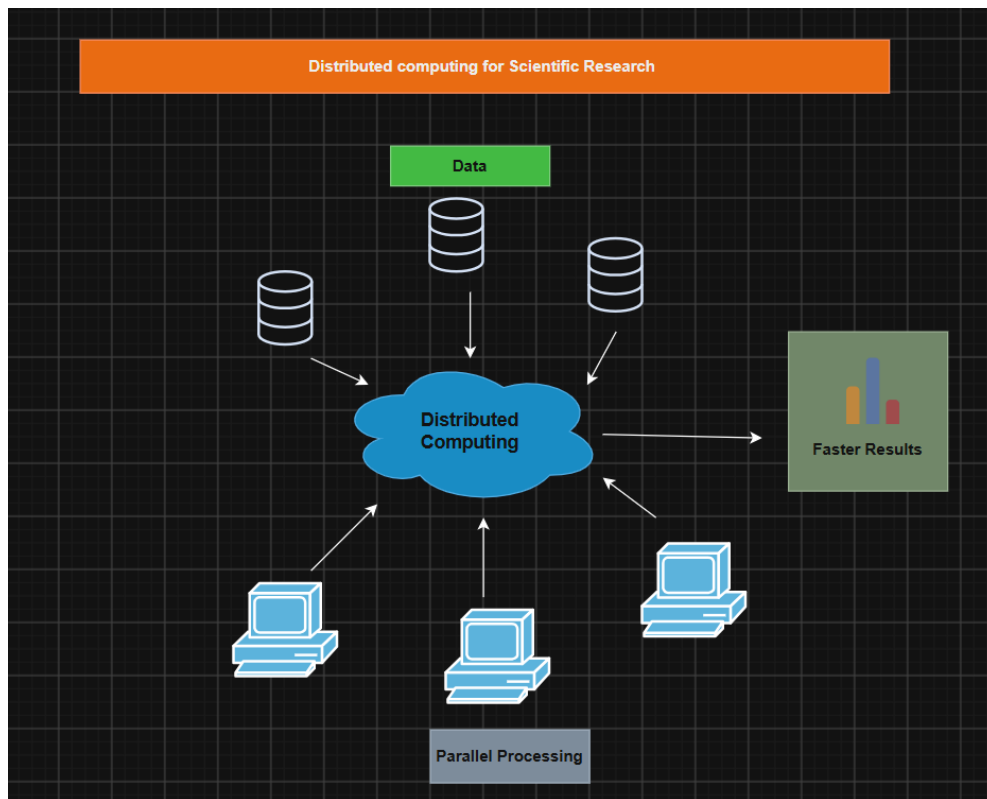
In a centralized system, one computer performs the complete processing. In distributed computing, the data is divided into smaller parts and distributed among many computers.

Parallel processing allows several computers to process different parts of the data at the same time. Therefore, processing becomes faster.

For example, weather research collects data from thousands of sensors. Instead of one computer processing all the information, different computers can process different sensor data simultaneously.

Technologies such as Hadoop and Apache Spark support distributed processing. Distributed computing also provides scalability because more computers can be added when the data increases.

Therefore, distributed computing is more suitable for large scientific datasets because it provides faster processing, scalability and better resource usage compared with a centralized system.



3. Design a fraud detection framework for a digital payment platform using big data analytics, and justify how streaming data processing and predictive models improve fraud prevention accuracy.

Ans:

Digital payment platforms process thousands or millions of transactions. Some transactions may be fraudulent, so a real-time fraud detection system is required.

The system collects information such as transaction amount, location, time, device and previous transaction history. Streaming technologies such as Apache Kafka can process transactions as they occur.

Big data processing checks every transaction and compares it with the customer's previous behaviour. Machine learning models can identify unusual patterns and calculate a fraud risk score.

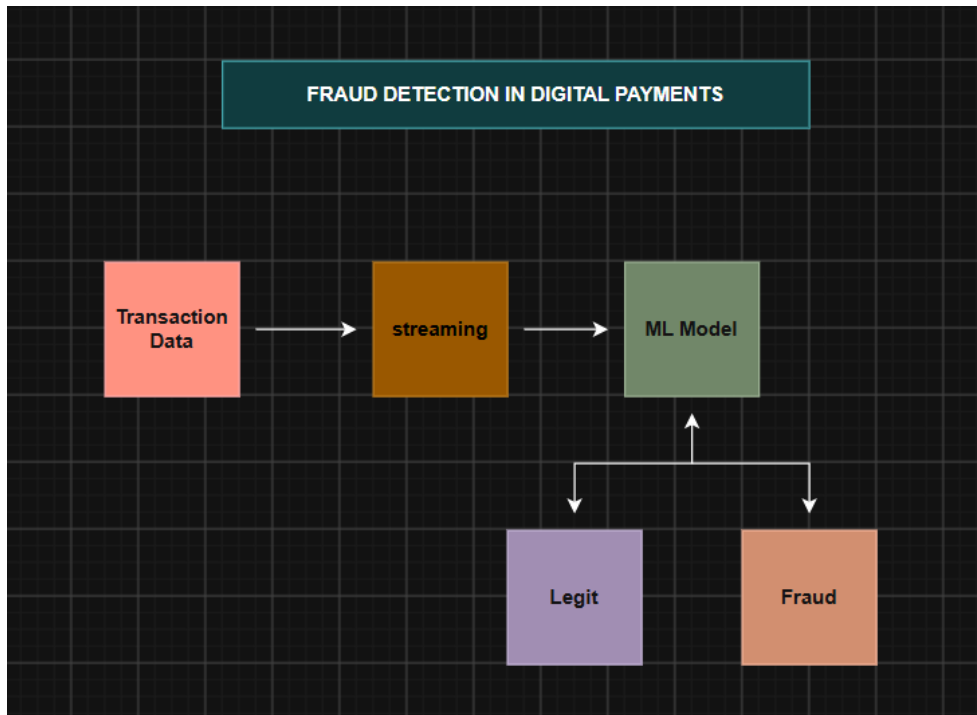
For example, if a customer normally makes payments of ₹500–₹2,000 but suddenly makes a ₹1 lakh transaction from a different location, the system can identify it as suspicious.

The process is:

Transaction → Streaming → Real-Time Processing → Fraud Model → Risk Score → Approve/Block

Real-time processing helps stop fraud quickly, while predictive models improve detection accuracy.

Thus, big data analytics helps digital payment companies reduce fraud, financial loss and security problems.



4. Assess the challenges of storing and processing multimedia data such as video, audio, and images in a big data environment, and recommend suitable technologies and storage mechanisms for efficient management.

Ans:

Organizations generate large amounts of videos, images and audio files. Managing this data is difficult because multimedia files are usually very large and require high storage and processing power.

One major problem is storage. High-quality videos can occupy several gigabytes. Another problem is processing because image and video analysis requires more computing power.

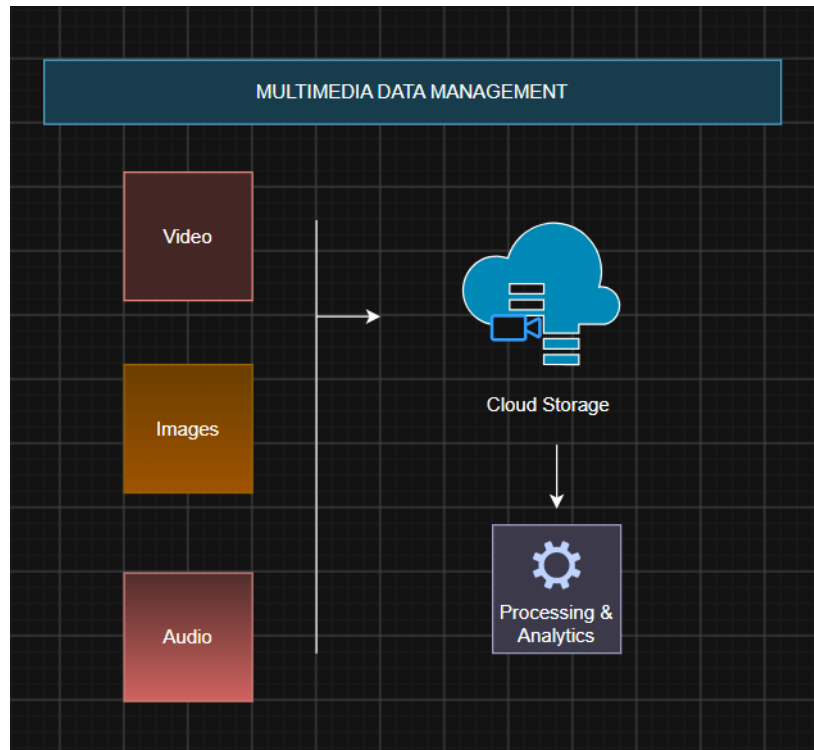
Distributed storage such as HDFS and cloud object storage can be used to store large multimedia files. Cloud storage can be increased easily when more data is generated.

Compression can reduce the size of files. Metadata such as file name, date, size and location can also be stored to make searching easier.

Tools such as Apache Spark can be used for large-scale processing, while AI and machine learning can be used for image and video analysis.

For example, a security company can store thousands of hours of CCTV video in cloud storage and use big data processing to identify important events.

Therefore, distributed storage, cloud storage, compression and parallel processing are useful for managing multimedia data.



5. Construct a disaster recovery and business continuity strategy for a cloud-based big data platform, and justify how redundancy, replication, and automated failover mechanisms ensure operational resilience.

Ans:

A cloud-based big data platform may contain important business information. System failure, cyber attacks or natural disasters can stop business operations. Therefore, disaster recovery and business continuity are important.

Redundancy means keeping additional servers and storage systems. If one server fails, another server can continue the service.

Replication means keeping copies of data in different servers or locations. This protects the organization from data loss.

Regular backups should also be taken. If data is accidentally deleted or damaged, the backup can be used to restore it.

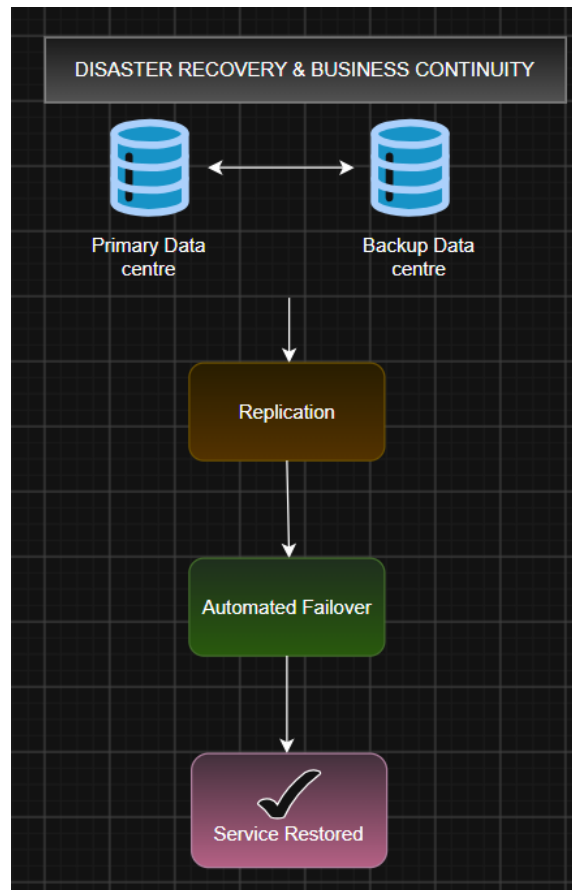
Automated fail over automatically transfers the application to a backup server when the main server fails. This reduces downtime.

The basic process is:

Main Server → Data Replication → Backup Server → Automatic Failover → Service Restored

For example, if a cloud data center becomes unavailable because of a disaster, the application can switch to another data center containing a copy of the data.

Therefore, redundancy, replication, backup and automated failover help maintain business operations and improve system reliability.



6. Analyze the impact of data quality issues such as inconsistency, duplication, and missing values in a large-scale analytics platform, and propose data governance strategies to improve analytical reliability.

Ans:

Data quality is important because incorrect data can produce incorrect analytical results. Large-scale systems commonly face problems such as **inconsistency, duplication and missing values**.

Inconsistency occurs when the same information is stored in different formats. For example, one system may store "Tamil Nadu" while another stores "TN".

Duplication occurs when the same customer is stored more than once. This can give incorrect customer counts.

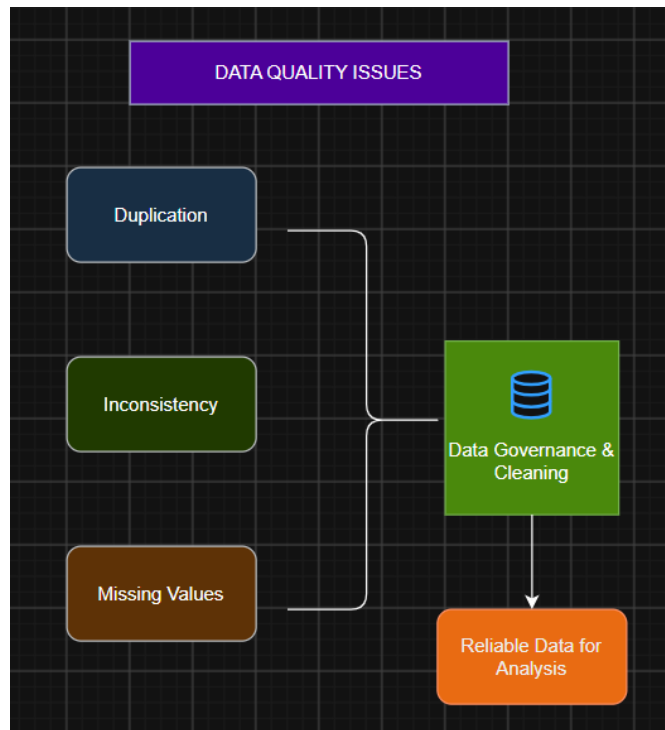
Missing values occur when important fields are empty. For example, a customer record may not contain an age or phone number.

These problems can be handled using data cleaning and validation. Duplicate records should be removed, formats should be standardized and missing values should be handled properly.

Data governance should also be used. It defines rules for collecting, storing and accessing data. Regular quality checks can measure accuracy, completeness and consistency.

For example, a retail company combining data from many stores should remove duplicate customer records before performing sales analysis.

Therefore, good data quality and proper governance improve the accuracy and reliability of big data analytics.



7. Design a scalable recommendation system for an online streaming service using big data technologies, and justify how user behavior analytics and distributed processing enhance recommendation accuracy and user engagement.

Ans:

Online streaming services have millions of users and a large amount of content. A recommendation system helps users find movies, songs or videos that they may like.

The system collects user behaviour such as watch history, search history, ratings, likes and viewing time. This information is stored in a big data platform.

Distributed processing tools such as Apache Spark can process data from millions of users. Machine learning models can identify user interests and recommend suitable content.

For example, if a user frequently watches action movies, the system can recommend similar movies.

Collaborative filtering can also compare users with similar interests and recommend content based on their behaviour.

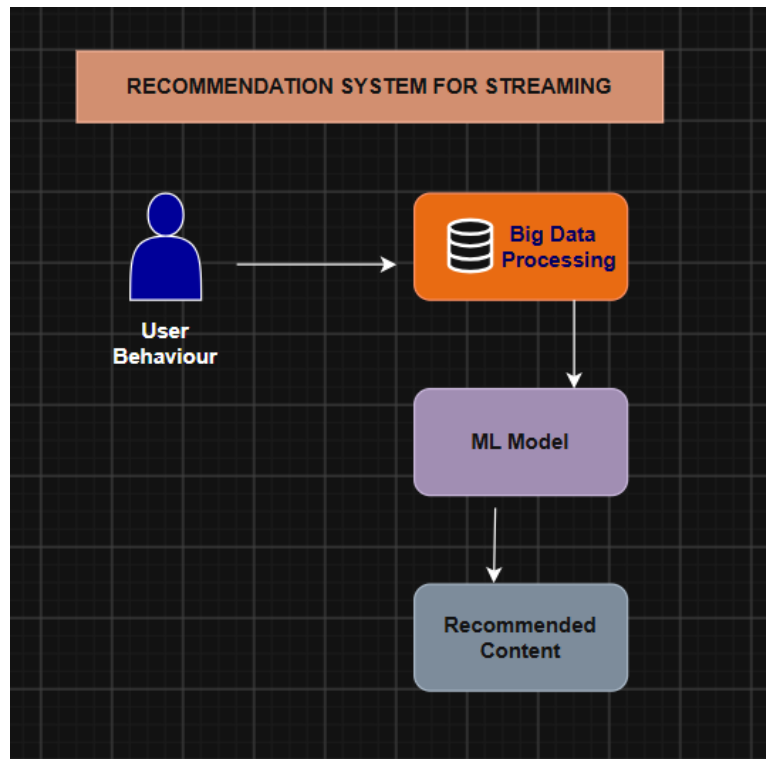
The basic flow is:

User Activity → Data Storage → Distributed Processing → Recommendation Model → Personalized Content

Real-time analytics can update recommendations when the user's interests change.

For example, Netflix can recommend movies and shows based on a user's previous viewing behaviour.

Therefore, big data and distributed processing improve recommendation accuracy, user engagement and customer satisfaction.



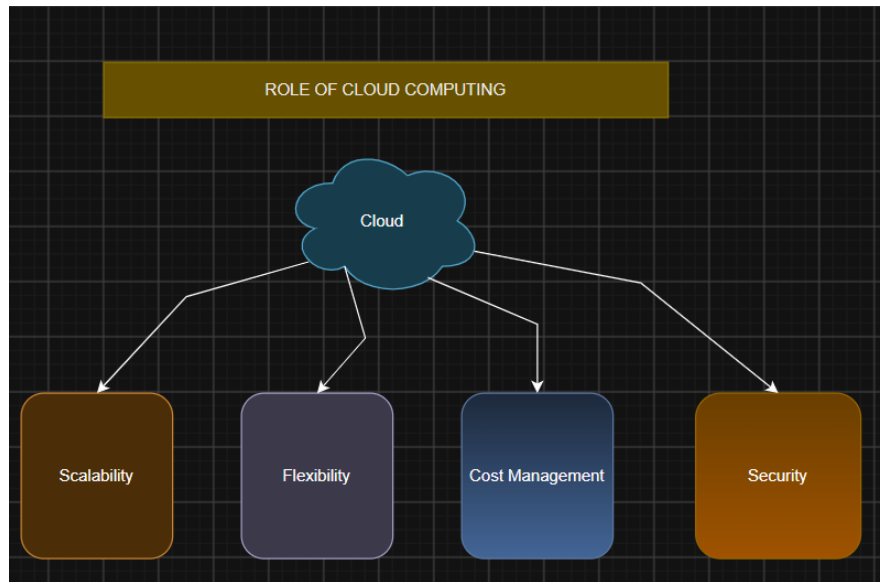
8. Evaluate the role of cloud computing in modern big data ecosystems, and critically examine its advantages and limitations in terms of scalability, cost management, flexibility, and security.

Ans:

Cloud computing is widely used in big data because big data applications require large amounts of storage and computing resources. One major advantage is scalability. Organizations can increase storage and computing power when the amount of data increases. Cloud computing also provides flexibility because organizations can use services such as cloud storage, databases, virtual machines and analytics platforms according to their needs. It can reduce initial costs because companies do not need to purchase large amounts of physical hardware. They can use a pay-as-you-use model.

However, cloud computing also has limitations. Poor resource management can increase costs. Security and privacy are concerns because sensitive data is stored on cloud infrastructure. Vendor lock-in can also make it difficult to move to another provider.

For example, a company can use cloud storage to store large datasets and Apache Spark services to analyze them without maintaining its own data center. Thus, cloud computing provides scalability, flexibility and easier resource management, but security, cost and vendor dependency must be properly managed.



9. Develop a real-time analytics framework for transportation and logistics management, and justify how big data technologies improve route optimization, fuel efficiency, and operational decision-making.

Ans:

Transportation companies generate data from GPS devices, vehicle sensors, traffic systems and delivery applications. Big data analytics can use this information to improve transportation and logistics.

GPS provides the location of vehicles, while traffic data provides information about road conditions. Streaming technologies can process this information in real time.

The system can identify traffic jams and suggest alternative routes. This reduces travel time and fuel consumption.

Big data can also analyze fuel usage and vehicle performance. If a vehicle uses more fuel than normal, the company can investigate the reason.

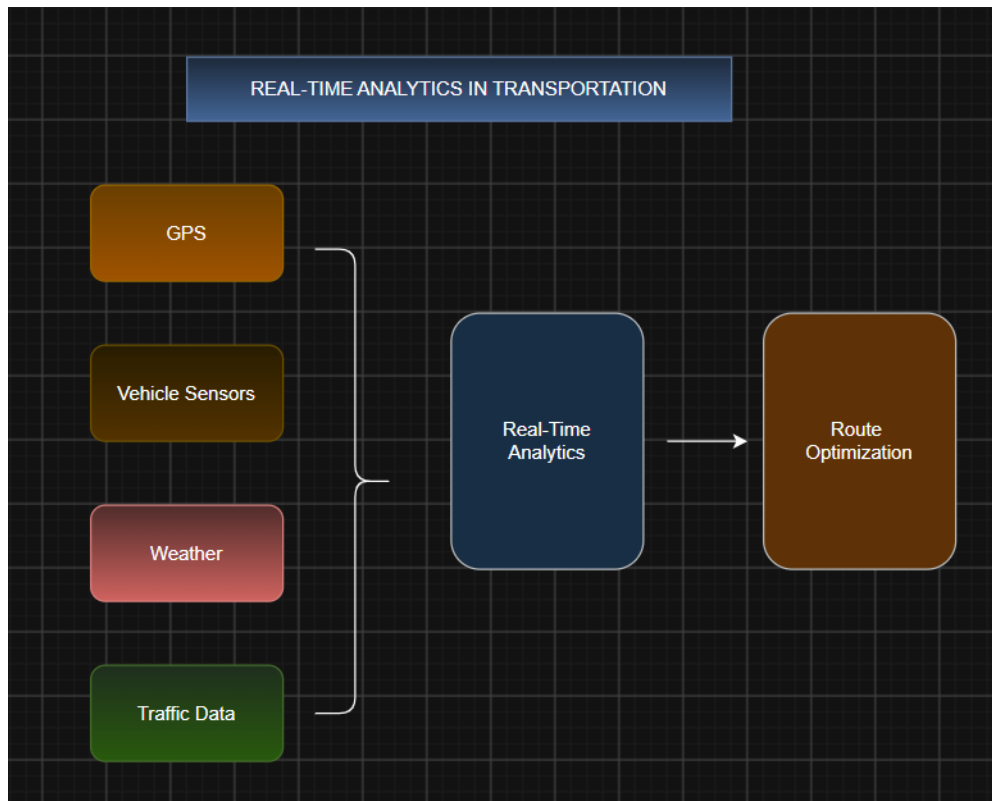
Predictive analytics can be used for vehicle maintenance. Sensor data can help identify possible problems before a vehicle breaks down.

The basic flow is:

GPS + Sensors → Streaming Data → Big Data Processing → Real-Time Analytics → Route Optimization

For example, a delivery company can change the route of a vehicle when heavy traffic is detected.

Therefore, big data improves route planning, fuel efficiency, delivery time and operational decision-making.



10. Critically analyze ethical and legal challenges associated with big data collection and analytics, and recommend organizational policies that ensure responsible data usage, regulatory compliance, and user trust.

Ans:

Big data provides many benefits, but organizations must use customer information responsibly. One major issue is privacy. Companies may collect personal information such as names, locations, purchase history and online activities.

Users should know what information is collected and why it is being used. Organizations should not collect unnecessary information without proper permission.

Data security is another important issue. Personal information can be stolen during cyber attacks. Encryption, authentication and access control should be used to protect the data.

There can also be bias in big data analytics. If the training data contains unfair information, machine learning models may also produce unfair results.

Organizations should follow applicable privacy and data protection regulations. They should create policies for user consent, data access, data retention and secure deletion.

For example, if an online shopping application collects a customer's location to provide nearby offers, the customer should be informed about this purpose and the information should be protected.

Therefore, responsible big data usage requires privacy, security, transparency, user consent and legal compliance. This helps organizations maintain customer trust while using data effectively.

